Reza Ramezanpour

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Education

Rice University — Houston, TX
PhD in Electrical and Computer Engineering — *Aug. 2023 – Present*
- Co-advised by Prof. Ashutosh Sabharwal and Prof. Santiago Segarra

Rice University — Houston, TX
Master of Science in Electrical & Computer Engineering — *Aug. 2023 – Sep. 2025*

Sharif University of Technology — Tehran, Iran
Bachelor of Science in Electrical Engineering — *Sep 2016 – Jan 2021*

Research Interests

Wireless Communications & Imaging: Resource allocation, ISAC, Multistatic Wireless Imaging, Scene Imaging and Reasoning
Computer Vision & Multi-modal Fusion: Camera–Radar Sensor Fusion, 3D Scene Reconstruction, Vision-Language Models
Graph Machine Learning: Graphon learning, Graph neural networks, Unsupervised representation learning
ML Theory: Statistical Learning Theory, Convex Optimization, Reinforcement Learning, Transformers & LLMs

Publications & Presentations

Ramezanpour, R., Segarra, S., and Sabharwal, A. (2024). Multiuser Fair Resource Allocation for Block-stationary Wireless Channels. *Asilomar Conference on Signals, Systems, and Computers (ACSSC)*. Link
↪ *Oral presentation at the Asilomar Conference, Monterey, CA, Oct 2024.*
↪*Poster presented at the NSF RINGS PI Meeting, Qualcomm HQ, San Diego, CA, Feb 2025.*

Ramezanpour, R., Tenorio, V. M., Marques, A. G., Sabharwal, A., and Segarra, S. (2025). A Few Moments Please: Scalable Graphon Learning via Moment Matching. *Advances in Neural Information Processing Systems (NeurIPS)*. Link
↪*Selected for poster presentation at NeurIPS, San Diego, CA, Dec 2025.*

Azizpour, A., **Ramezanpour, R.**, Sabharwal, A., and Segarra, S. (2026). From Moments to Models: Graphon Mixture-Aware Mixup and Contrastive Learning. *International Conference on Machine Learning (ICML)*. Link

Experience

University of California, San Diego — San Diego, CA

Visiting Graduate Student — *Summer 2026 – Present*
- Capture multistatic radar imaging data and apply computer vision techniques to enhance 3D reconstruction in the mmWave band.

Rice University — Houston, TX

Research Assistant, Wireless and Data Science Group — *Spring 2024 – Present*
- Develop novel machine learning and optimization techniques for Wireless and Graph ML applications under the supervision of Prof. Ashutosh Sabharwal and Prof. Santiago Segarra.

Teaching Assistant, Intro to Machine Learning — *Fall 2025*
- Lead weekly help sessions and grade assignments for a graduate-level Machine Learning course, mentoring 50+ students on advanced topics.

Teaching Assistant, Information Theory — *Fall 2024*
- Led weekly help sessions and graded all homeworks, providing detailed feedback to students.

Sharif University of Technology — Tehran, Iran

Research Assistant, Data Science Group — *Aug 2020 – Aug 2021*

- Investigated machine learning applications for traffic prediction and database management under the supervision of Dr. Iman Gholampour.

Teaching Assistant *Sep 2016 – Jan 2021*

- Courses: *Parallel Computing, Numerical Calculation, Computer Architecture, Object-Oriented Programming.*
- Developed and graded theoretical and computational assignments; mentored students during lab sessions on coding and circuit simulation.

Research Projects

Graph Learning-based Fair Resource Allocation for Multi-User Wireless Channels Master Thesis

- Developed a novel, GNN-based framework to find a polynomial-time solution to the NP-hard problem of Proportional Fair resource allocation. My approach combines a GNN-based power allocator with an efficient user pruning strategy. Also developed an LLM-based user selection model to investigate the performance of general experts for this task. The framework achieves high throughput and fairness while remaining robust to distributional shifts. Soon to be submitted to the *IEEE Transactions on Machine Learning in Communications and Networking.*

Multiuser Fair Resource Allocation for Block-stationary Wireless Channels ACSSC 2024

- We propose an algorithm that leverages the block-stationarity of the environment to achieve a gain over stationary solvers. Since this problem is NP-hard, we also introduce two complexity reduction techniques, one from the mathematical domain and another from deep learning, to obtain a polynomial-time solution. A poster of this work is presented in RINGS PI3.

A Few Moments Please: Scalable Graphon Learning via Moment Matching NeurIPS 2025

- Designed MomentNet, a novel, polynomial-time graphon estimator that resolves the critical scalability bottleneck of prior methods. It uses an Implicit Neural Representation (INR) for direct graphon recovery via moment matching, outperforming state-of-the-art estimators in 75% of benchmarks.
- Supported the method with a theoretical guarantee that bounds the estimation error and developed MomentMixup, a new data augmentation technique that improves graph classification by interpolating in the moment space.

From Moments to Models: Graphon Mixture-Aware Mixup and Contrastive Learning ICML 2026

- Proposed a framework that models graph data as a mixture of graphons to account for hidden population structures, using graph moments to disentangle the components. This enabled two novel methods, GMAM (a graphon-aware Mixup) and MGCL (a model-aware contrastive learning approach), that set new state-of-the-art results across numerous supervised and unsupervised benchmarks.

Sensor-Aware Camera–Radar Radiance Fields for 3D Scene Reconstruction *Preprint*

- Developed a camera–radar fusion framework that learns a 3D scene representation in which each sensor supervises only the geometry it can reliably measure, combining the camera's shape and semantic cues with radar's metric range for more accurate 3D reconstruction.

BiNorm: Bistatic Normal Cues for Sparse mmWave Surface Reconstruction *Preprint*

- Developed a multistatic mmWave imaging approach that estimates surface normals from distributed bistatic radar nodes and uses them as sparse constraints in an implicit neural surface representation, yielding more coherent object reconstructions from limited RF measurements.

VLM-Guided Wireless Imaging for High-Fidelity 3D Reconstruction Ongoing

- Designing a wireless imaging reasoner that enables question-based reasoning about the environment and supports cooperative multistatic sensing across distributed nodes. The framework uses Vision-Language Models (VLMs) to reason over wireless imaging observations, answer questions about the surrounding environment, and combine complementary information from distributed sensing nodes for richer scene understanding.

OTHER PROJECTS

Classification of Synaptic Connectivity in an Imbalanced Domain Statistical Learning

- Addressed the challenge of severe class imbalance in synaptic connectivity prediction by designing a custom Weighted Vote Classifier. Operating on features extracted from raw neuronal signals, the proposed model significantly outperformed standard resampling (SMOTE) and weighting methods, demonstrating a more robust approach for imbalanced neuroscientific datasets.

Scalable Traffic Pattern Analysis and Route Optimization using Big Data *Big Data Analysis*

- Designed a scalable, multi-stage framework in Apache Spark to analyze large-scale traffic data for congestion analysis and route optimization. The framework integrates graph analytics (PageRank, community detection) to identify critical congestion hubs and employs a hybrid machine learning model of frequent pattern mining (FP-Growth) and collaborative filtering (ALS) to generate personalized route recommendations.

Selected Courses

Machine Learning (A)

Information Theory (A)

Big Data Analysis (A)

Parallel Programming and Architectures (A)

Reinforcement Learning (A)

Statistical Learning (A+)

Efficient Implementation of Neural Networks (A)

Intro to Vision (A) – Diffusion Models, Flow Matching, Vision-Language Models (VLMs)

Technical Skills

Programming Languages: Python, C/C++, Java, R, SQL
Machine Learning Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, PyTorch Geometric (PyG)
Data Science & Visualization: NumPy, Pandas, Matplotlib, Seaborn, Scipy
ML Concepts: Deep Learning (Graph Neural Networks, Transformers), Supervised & Unsupervised Learning, Feature Engineering, Model Tuning & Evaluation, Reinforcement Learning, Large Language Model prompt engineering, VLM fine-tuning, NLP, Computer Vision
High Performance Computing: CUDA, OpenMP, Multithreading, Parallel Algorithm Design
Databases: Relational (MySQL, PostgreSQL), NoSQL (MongoDB)
Developer Tools & Practices: Git, Docker, Linux/Unix Environment, Bash Scripting, LaTeX